

BOB-140 “ CM-HEALTHCHECK-COVERS-SERVED-PORTS upstreamed into the constitution: closure evidence

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What BOB-140 asked for

scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh (landed with BOB-138) implemented a rule every project under this constitution needs “ a container health check MUST cover every port its service serves, because a check probing a subset asserts a proxy signal instead of the real condition (Â§11.4.201). Its detection logic already carried zero boba literals, so per Â§11.4.177 / Â§11.4.28 / Â§11.4.74 it belongs in the constitution and must be consumed BY REFERENCE, with boba shipping a thin delegator holding only scope DATA (Â§11.4.35).

The deferral was deliberate and tracked, not an oversight: a concurrent agent was editing constitution/scripts/gates/ at the time, and two writers in one submodule directory risks losing work (Â§11.4.197). That directory was clean at constitution HEAD 997d48f when this work began.

What landed

File	Role
constitution/scripts/gates/ cm_healthcheck_covers_served_ports.sh	NEW “ the universal detection ENGINE
constitution/scripts/gates/ cm_healthcheck_covers_served_ports_mutation_test.sh	NEW “ the paired Â§1.1 mutation test
scripts/pre_build/ check_cm_healthcheck_covers_served_ports.sh	CONVERTED “ thin delegator, scope DATA only

config/served_ports.yaml needed no change: the manifest schema the engine consumes is exactly the one boba already declared.

The engine’s interface is --compose <file> --manifest <file> [--quiet], with HEALTHCHECK_PORTS_COMPOSE / HEALTHCHECK_PORTS_MANIFEST /

HEALTHCHECK_PORTS_PYTHON env equivalents. It carries **no** project literal and refuses (exit 2) rather than guessing a consumerâ€™s paths.

The two properties that had to survive the move

Both are load-bearing. Each is now a property of the ENGINE, asserted by the engineâ€™s own mutation test rather than by prose:

1. **Zero services checked â†’ FAIL.** A quiet zero from a blind instrument is indistinguishable from a clean tree (Â§11.4.201(6)). Two independent routes are covered: an empty manifest, and a manifest whose every serves list is empty (an implementation counting skipped entries as â€œcheckedâ€ would pass the second).
2. **No python with PyYAML â†’ FAIL, never SKIP.** Without a parser the gate cannot assert anything; Â§11.4.201(4)â€™s conservative-safe outcome is to refuse and say so honestly.

Neither degraded. Evidence for both is pasted below (sections D and E), taken through the delegator on bobaâ€™s real tree.

A. Output equivalence with the pre-move implementation

Both spellings run against the same real compose file and manifest. The old implementation is `git show HEAD:` of the pre-conversion script, run with explicit paths (a copy outside the repo cannot derive bobaâ€™s PROJECT_ROOT).

```
old rc=0
new rc=0
--- stdout diff (empty == byte-identical) ---
STDOUT IDENTICAL
--- stderr diff ---
STDERR IDENTICAL
--- md5 ---
bd38e61fe453938dd28fb971c927c5cb /tmp/out_old.txt
bd38e61fe453938dd28fb971c927c5cb /tmp/out_new.txt
```

The invocation contract is therefore preserved exactly: same path, same optional positional [COMPOSE_FILE] [MANIFEST], same exit codes, same PASS: / FAIL: line shapes, and the PASS line is still **last** â€” which matters, because pre-build invariant 44 reads it with `tail -n1`.

B. Delegator on bobaâ€™s real tree

```
$ bash scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh
ok qbittorrent: healthcheck covers all served ports [7185]
ok jackett: healthcheck covers all served ports [9117]
```

```

ok download-proxy: healthcheck covers all served ports [7186, 7187]
ok qbittorrent-proxy-go: healthcheck covers all served ports [7187]
ok boba-jackett: healthcheck covers all served ports [7189]
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
rc=0

```

C. Paired “1.1 mutation test” both directions

13 fixtures, 11 FAIL-on-mutation and 2 PASS-on-clean negative controls.

```

=====
1.1 paired-mutation meta-test for CM-HEALTHCHECK-COVERS-SERVED-PORTS
anchors: 11.4.201 (guard asserts the real condition) / 11.4.254
fixtures under: /tmp/.private/milosvasic/hcports_mut.thRv7J
=====
âœ… META OK: BOB-138 literal shape (serves 7186+7187, probes only 7186)
âœ… META OK: BOB-138 fixed shape (health check probes BOTH served ports)
âœ… META OK: declared service with NO healthcheck at all “ gate correct
âœ… META OK: substring near-miss (serves 7187, probes 71870 “ must NOT
âœ… META OK: UNDECLARED compose service with a healthcheck + ports “ g
âœ… META OK: STALE manifest entry (declared but absent from the compose
âœ… META OK: ZERO services checked (empty manifest) “ blind, not clean
âœ… META OK: ZERO services checked (all 'serves' lists empty) “ blind,
âœ… META OK: no python with PyYAML “ BLIND gate must FAIL, not skip âœ…
âœ… META OK: missing compose file (blind input) “ gate correctly FAILE
âœ… META OK: missing manifest file (blind input) “ gate correctly FAIL
âœ… META OK: negative control: every service fully covered (list + strin
âœ… META OK: negative control: port-less worker absent from manifest is
=====
âœ… META PASS “ CM-HEALTHCHECK-COVERS-SERVED-PORTS FAILs-on-mutation AND
META rc=0

```

The two negative controls exist because 11.4.201(1) makes a false-positive refusal a FAIL-bluff exactly as a false pass is a PASS-bluff. Control 1 is a multi-service fixture where every health check legitimately covers all its served ports, exercising both test: spellings (list form and bare string form) plus a genuine multi-port probe. Control 2 pins the undeclared-service rule’s boundary: a port-less worker absent from the manifest publishes nothing and must not be reported.

D. Zero-services case, through the delegator

```

$ bash scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh "$T/c
CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL “ checked 0 services; the gate
rc=1 <-- MUST be 1 (blind, not clean)

```

E. Missing-PyYAML case, through the delegator

The delegator's interpreter candidates include an **absolute** `.venv/bin/python` path, so PATH shadowing alone cannot reach it. The fixture therefore builds a throw-away project root containing the real delegator, the real constitution (by symlink) and byte-identical copies of the real compose file and manifest " but no `.venv`, so every candidate is PATH-reachable and shadowable. The sanity run proves the fixture PASSES with a real interpreter first, so the subsequent FAIL is attributable to the interpreter and nothing else.

```
Sanity: this throw-away root PASSES with a real interpreter "
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
rc=0
```

Now shadow `python3/python` with interpreter-shaped executables that have no ONLY the interpreter changes " the compose file and manifest are byte-id CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL " no python interpreter with Py tried: `/tmp/.private/milosvasic/tmp.JMcPyCC3d7/root/.venv/bin/python` py the gate cannot parse the compose file, so it cannot assert anything. This is a BLIND gate, not a clean tree (11.4.201(6)).
rc=1 <-- MUST be 1: a BLIND gate FAILs, it never skips and never passes

F. Real mutation on boba's own compose file, through the delegator

Reverting the download-proxy health check to its exact pre-BOB-138 state " one line changed, nothing else " must be refused.

```
--- the ONLY difference from the real compose file ---
230c230
<      test: ["CMD-SHELL", "python -c \"import urllib.request as u; u.url
---
>      test: ["CMD-SHELL", "python -c \"import urllib.request as u; u.url

--- delegator against the MUTATED compose (real manifest, unchanged) ---
CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL
- download-proxy: serves [7186, 7187] but its healthcheck probes none of
  healthcheck: CMD-SHELL python -c "import urllib.request as u; u.url
  -> a dead port in [7187] reports HEALTHY forever (11.4.201 proxy s
ok qbittorrent: healthcheck covers all served ports [7185]
ok jaxett: healthcheck covers all served ports [9117]
ok qbittorrent-proxy-go: healthcheck covers all served ports [7187]
ok boba-jaxett: healthcheck covers all served ports [7189]
rc=1 <-- MUST be 1

--- G. paired control: same command, UNMUTATED compose ---
ok qbittorrent: healthcheck covers all served ports [7185]
ok jaxett: healthcheck covers all served ports [9117]
```

```

ok download-proxy: healthcheck covers all served ports [7186, 7187]
ok qbittorrent-proxy-go: healthcheck covers all served ports [7187]
ok boba-jackett: healthcheck covers all served ports [7189]
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
rc=0 <-- MUST be 0

```

H. No duplicated detection logic (Â§11.4.177)

The delegatorâ€™s complete executable body â€” 19 lines, comments and blanks stripped â€” is engine resolution, scope DATA and one exec:

```

1  set -euo pipefail
2  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
3  PROJECT_ROOT="$(cd "${SCRIPT_DIR}/../.." && pwd)"
4  CONST_GATE="${PROJECT_ROOT}/constitution/scripts/gates/cm_healthch
5  if [[ "${1:-}" == "--help" || "${1:-}" == "-h" ]]; then
6      sed -n '2,80p' "${BASH_SOURCE[0]}" | sed 's/^# \{0,1\}//'
7      exit 0
8  fi
9  if [[ ! -f "$CONST_GATE" ]]; then
10     echo "CM-HEALTHCHECK-COVERS-SERVED-PORTS: ERROR â€” shared gat
11     echo "    (Â§11.4.177 â€” this gate is consumed by reference; ru
12     echo "    'git submodule update --init constitution' to restore
13     exit 2
14  fi
15  COMPOSE_FILE="${1:-${PROJECT_ROOT}/docker-compose.yml}"
16  MANIFEST="${2:-${PROJECT_ROOT}/config/served_ports.yaml}"
17  PY_CANDIDATES="${PYTHON_BIN:-} ${PROJECT_ROOT}/.venv/bin/python py
18  exec env HEALTHCHECK_PORTS_PYTHON="$PY_CANDIDATES" \
19      bash "$CONST_GATE" --compose "$COMPOSE_FILE" --manifest "$MANI

```

Detection-primitive census over executable lines only:

primitive	delegator engine	
yaml	1	5
safe_load	0	2
re.search	0	1
healthcheck	1	7
serves	0	6
PYEOF	0	2
UNDECLARED	0	1
findings	0	7
checked	0	5

The delegatorâ€™s two non-zero counts are **file paths, not logic**: config/served_ports.yaml on line 16 and the engine filename on lines 4 and 10. Every detection primitive lives in exactly one place.

I. Invariant 44 wiring intact

scripts/pre_build_verification.sh was **not** edited. Its exact code path, replayed:

```
CONST_GATE_TIMEOUT = 300s
  æ... PASS CM-HEALTHCHECK-COVERS-SERVED-PORTS: CM-HEALTHCHECK-COVERS-SERV
    (747ms, well inside the 300s budget)
```

The doubled gate name is pre-existing formatting in invariant 44's pass call, not a regression æ" it prefixes the label onto tail -n1 of the gate's own output, and the gate's output has always begun with its own name.

Syntax checks

```
engine bash -n OK
mutation-test bash -n OK
delegator bash -n OK
```

Honest boundary (Æ11.4.6)

Stated as facts, not assumed:

- **PYTHON_BIN semantics changed, deliberately and for the stronger.** The old script trusted a set PYTHON_BIN on sight, with no import yaml probe. It is now the first candidate but is probed like any other, so a PYTHON_BIN lacking PyYAML falls through to the next candidate instead of failing confusingly downstream (Æ11.4.201(11): probe the artifact through its real invocation path). This is documented in the delegator header, not silent.
- **A new exit code 2 exists** for a missing engine or an engine usage error. Invariant 44 treats any non-zero, non-124 status as FAIL, so a missing constitution submodule blocks the build rather than passing æ" the correct direction. Exit codes 0 and 1 are unchanged.
- **Docker-compose parsing is unchanged**, including the shape it does not handle: the engine reads services.<name>.healthcheck.test and services.<name>.ports from a single compose file. It does not resolve extends, multi-file overlays, or profiles. That limit existed before this move and is not newly introduced; it is not currently exercised by boba.
- **The mutation test needs PyYAML to run.** If absent it exits 2 (environment error), never 0 æ" an unseeing meta-test must not report success.
- **docs/scripts/check_cm_healthcheck_covers_served_ports.md does not exist.** Æ11.4.18 requires a companion user guide for every shell script. This gap pre-dates BOB-140 (the pre-move script's header cross-referenced the document as though it existed). The delegator's header now states the gap honestly instead of citing a non-existent file; boba has no CM-SCRIPT-DOCS-SYNC invariant wired in scripts/pre_build_verification.sh, so nothing mechanical catches it. **This remains open and is not claimed closed** æ" docs/scripts/

is outside this work item's file ownership, so it was not created here.

- **Nothing was committed or pushed** in either repository, and no container was restarted, per this work item's scope.

Verdict

CM-HEALTHCHECK-COVERS-SERVED-PORTS now lives in the constitution as a project-agnostic engine with a paired 1.1 mutation test proving both polarities; boba consumes it by reference through a 19-line delegator carrying only scope DATA; the invocation contract and output are byte-identical to the pre-move implementation; and both load-bearing FAIL properties "zero services checked, and no usable YAML parser" survive the move, proven end-to-end through the delegator.